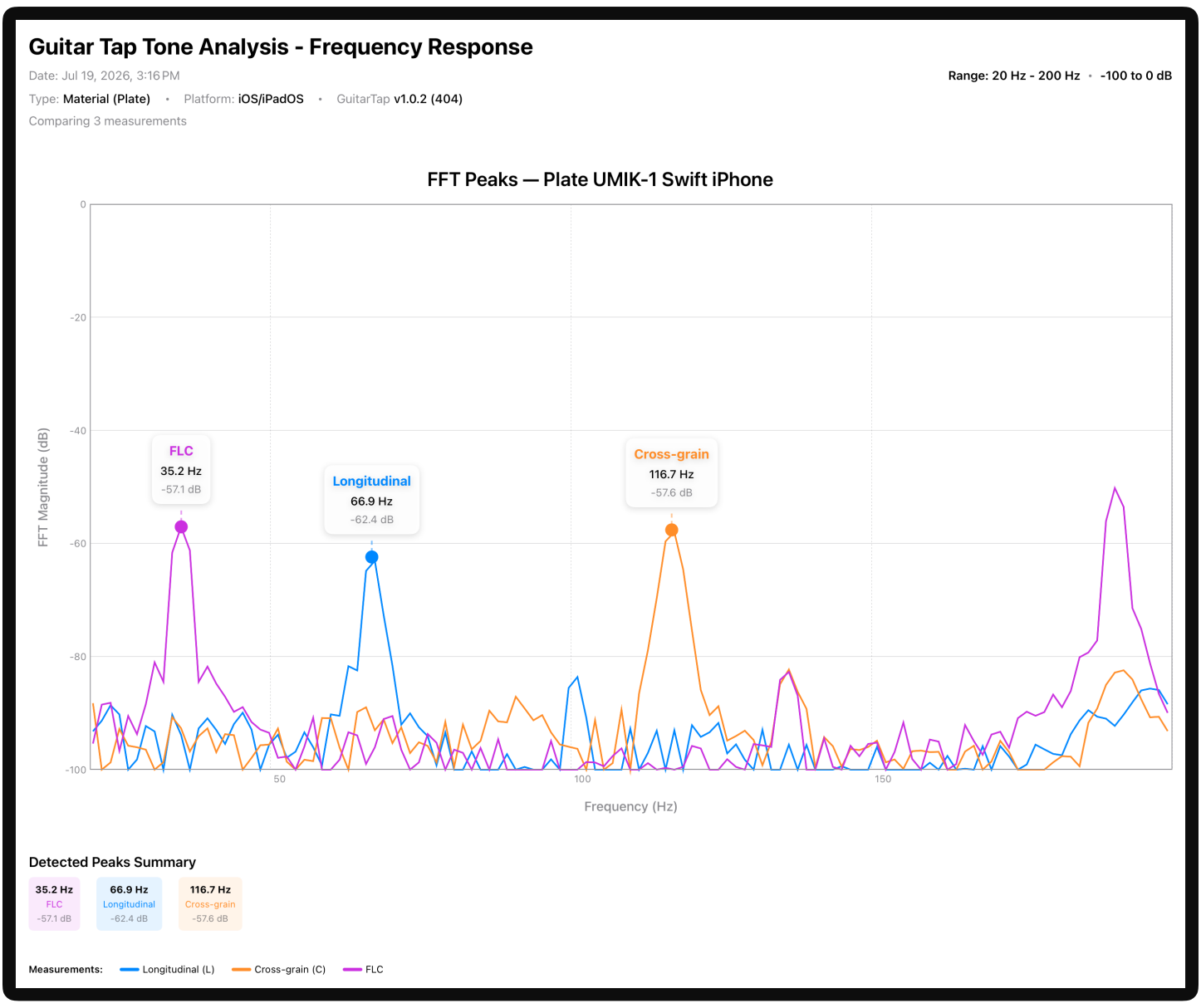
GuitarTap

Tap Tone Analysis Report

Jul 19, 2026, 3:16 PM

Measurement Name:	Plate UMIK-1 Swift iPhone
Type:	Material (Plate)
Frequency Range:	20.0 Hz – 200.0 Hz
Microphone:	Umik-1 Gain: 18dB · uncalibrated

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
35.2 Hz	-57.1 dB	C#1	12.0	FLC (Diagonal)
66.9 Hz	-62.4 dB	C2	15.3	Longitudinal (L)
116.7 Hz	-57.6 dB	A#2	26.7	Cross-grain (C)

Gore Target Thickness

3.52 mm

Body: 490 × 368 mm · f_vs = 75 (Steel String Top)

GLC (Shear Modulus): 0.337 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.2 mm** Thickness: **4.85 mm**

Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **66.9 Hz** fC: **116.7 Hz** fLC: **35.2 Hz**

Speed of Sound (L): **4170 m/s**

Speed of Sound (C): **1136 m/s**

Young's Modulus (L): **6.08 GPa**

Young's Modulus (C): **0.45 GPa**

GLC (Shear Modulus): **0.337 GPa**

Specific Modulus (L): **17.4** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **11.9**

Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
typical: 0.04–0.08

Long/Cross Ratio: **13.5**
typical: 12–25

Overall Quality: **Good**

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.